

# TAWHID ATHER

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## EDUCATION

### B.S. Computer Science, Minor in Biomedical Engineering

May 2026

University of Colorado Boulder

GPA: 3.9

**Coursework:** Algorithms, Data Structures, Software Development Methods & Tools, Database Systems, Machine Learning

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, C#, Java, C++, Bash, HTML5, CSS3/SCSS

**Web & Back End:** Angular, React, Node.js, Express, Flask, FastAPI, ASP.NET Core, Entity Framework, REST APIs, OAuth 2.0

**Data & AI:** PostgreSQL, MySQL, SQLite, Snowflake, SQLAlchemy, ETL, LangChain, AWS Bedrock, scikit-learn, Pandas

**DevOps & Tools:** Docker, GitHub Actions CI/CD, Ansible, Linux, Git, Jira, Postman, pytest, Mocha, Agile/Scrum

## EXPERIENCE

### Software Developer Intern

May 2025 – May 2026

Laboratory for Atmospheric and Space Physics (LASP), CU Boulder

Boulder, CO

- Built Python/Flask backend services and middleware supporting CI/CD pipelines for infrastructure used daily by 50+ researchers
- Built a full-stack Angular/TypeScript URL shortener letting scientists and engineers manage links to research data and papers
- Automated configuration and deployment of the web app and an SMTP notification service on Linux production servers with Ansible
- Built an automated email service that monitors user login activity and warns users who pass a set inactivity timeframe
- Wrote Python and Bash CLI tools and a custom log parser that automated deployments, maintenance, error detection and status reporting
- Provisioned Ubuntu/Debian servers and resolved production issues through root-cause analysis with science and engineering teams
- Secured applications with OAuth 2.0 and role-based access control; tested and documented REST APIs with Postman
- Worked in an Agile team using Git/Bitbucket and Jira, with daily stand-ups, sprint planning and code reviews

### Senior Resident Advisor

Aug 2023 – May 2026

University of Colorado Boulder

Boulder, CO

- Supervised and mentored 18 Resident Advisors across two halls serving 600+ students; led staff meetings, trainings and scheduling
- Handled confidential student records under FERPA, applying data privacy practices to sensitive personal information
- Mediated resident conflicts and gave crisis support, communicating clearly with students, staff and administrators

### President, Muslim Student Association, CU Boulder

Aug 2022 – May 2026

- Grew event participation 50% through outreach; ran year-round events with 200+ attendees and managed budgets and partnerships

## PROJECTS

### Festo Agentic Frontend Generator (Capstone, team of 6) | LangChain, AWS Bedrock, React

Oct 2025 – May 2026

- Architected the LangChain agent on AWS Bedrock that generates React/TypeScript/Tailwind UIs from Festo's component library
- Built a human-in-the-loop validation service with approval breakpoints so engineers vet LLM output before code generation proceeds
- Integrated validation into a live preview viewer and used prompt engineering to enforce consistent UI patterns across pages
- Added Docker health checks, logging and restart policies to keep the agent service, validation API and live preview available

### Budget Buddy | Python, FastAPI, SQLAlchemy, SQLite, Expo/React Native, GitHub Actions

2026 – Present

- Building a local-first finance app: FastAPI backend, Alembic migrations, accounts and transactions APIs and a mobile dashboard
- Implemented encryption at rest (OS-keyring master key, encrypted columns and files); 185+ pytest tests with server and mobile CI
- Run development as small pull requests drafted by specialized AI agents, reviewing every change and recording architecture decisions

### NER Tenets – Rent Data ETL Pipeline | Python, SQL, PostgreSQL, Snowflake, Docker

2026

- Built a Dockerized ETL pipeline loading HUD and Census API data into raw, staging and analytics schemas and a net effective rent view
- Ran identical pipeline code against a local Postgres sandbox and Snowflake, with shell scripts for end-to-end runs and ad hoc queries

### Spot U | Node.js, Express, PostgreSQL, Handlebars, Docker

Jan 2025 – May 2025

- Led a team of 5 building REST APIs for auth, playlists and social features with Spotify OAuth for 100+ test users; 85% Mocha coverage
- Designed and tuned the PostgreSQL schema, indexes and queries; Docker cut environment setup time 75%
- Built a responsive HTML/CSS/JavaScript interface with Handlebars templates, refined through iterative user testing

### BallXPit Packing Solver | Python, Google OR-Tools CP-SAT

2026

- Built a CP-SAT solver packing 28 polyomino buildings into two 7x7 zones, with a reachability constraint keeping a walkable corridor

### CRM & Payroll API | C#, .NET 10, ASP.NET Core, Entity Framework Core, PostgreSQL

2026

- Built a REST web API with contact and payroll controllers, a repository layer and Entity Framework Core migrations on PostgreSQL

### Developer Portfolio | Angular, TypeScript, SCSS, GitHub Actions

2026

- Built a responsive Angular portfolio with standalone components and SCSS design tokens, deployed to GitHub Pages through CI

### Chicago Housing Price Prediction | Python, scikit-learn, Pandas, NumPy

Aug 2024 – Dec 2024

- Built a regularized regression model (Lasso, Ridge) with feature engineering and cross-validation, reaching an R-squared of 0.82

### Caution: Slick Floors (Big Mode Game Jam 2026) | Unity, C#

Jan 2026

- Built a momentum-based 2D physics platformer with event-driven animation; ranked #214 in Theme among 647 entries